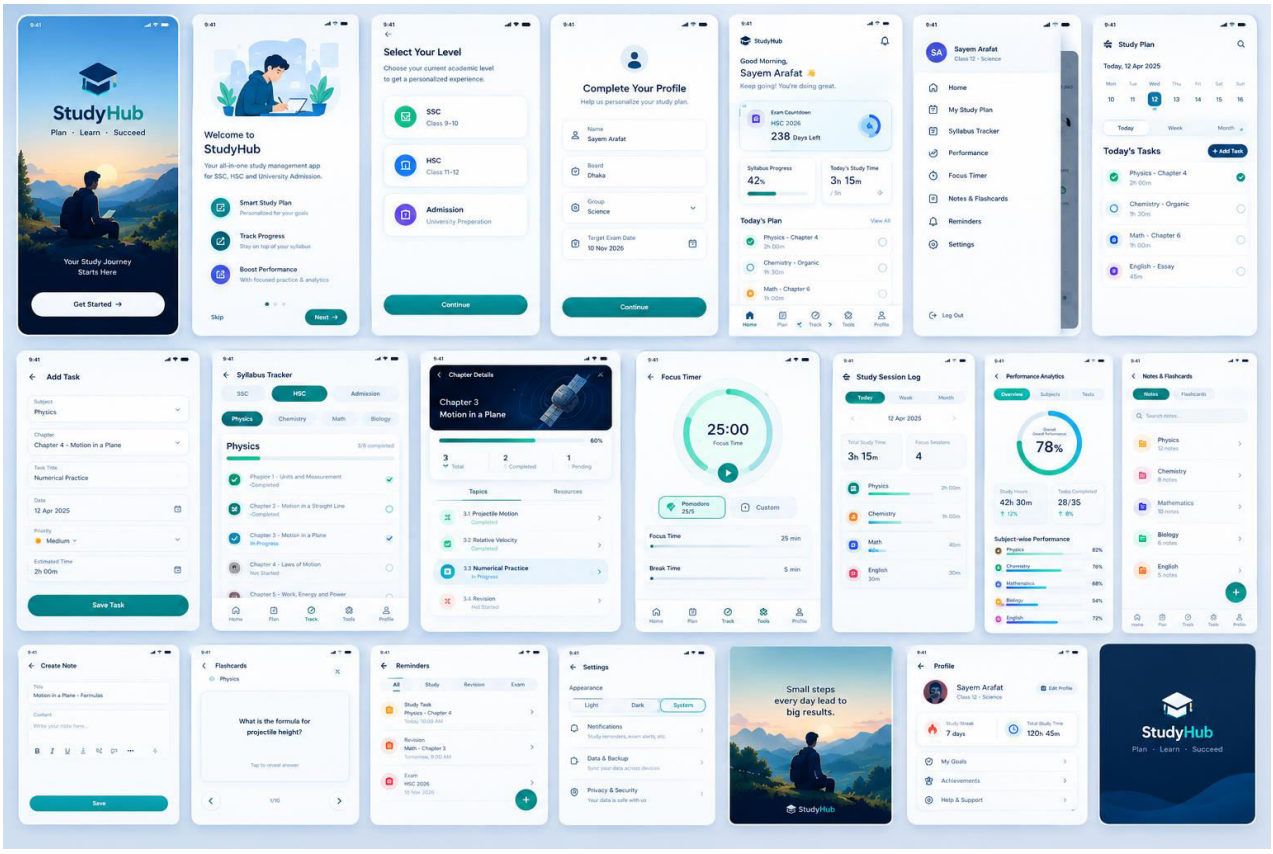
StudyHub

MASTER PRODUCT REQUIREMENTS DOCUMENT

Android-first study management app for Bangladesh's SSC, HSC and university admission students

Version 1.0 • Product + UX + Engineering specification • 24 October 2026



Visual reference: StudyHub master UI montage. Treat this as the direction and screen inventory, not permission to reproduce inconsistencies or legibility defects in the collage.

Document control

Field	Details
Product name	StudyHub
Platform	Android first; architecture should not block future web/iOS expansion
Primary market	Bangladesh
Audience	SSC, HSC and university admission candidates
Document owner	Product founder
Status	Build-ready product specification; validate curriculum and store-policy details before release
Primary design reference	The StudyHub light-theme UI montage generated in this conversation

How to use this PRD

This document is the single source of truth for the first implementation in Google AI Studio. Use it together with the attached visual reference image. The reference defines the intended product feel and broad screen inventory. This PRD defines exact behavior, consistency rules, data requirements, and acceptance criteria. Where a collage contains small text, approximate dates, placeholder scores, or inconsistent details, this document overrides those artifacts.

Implementation rule: do not build only a static mockup. Every visible primary control must perform a real, testable action or be explicitly marked as a disabled/not-yet-available feature. Do not silently invent production data.

1. Executive summary

StudyHub is a calm, premium, mobile-first study management app that helps Bangladeshi students organize their study life. It brings daily planning, syllabus tracking, focus sessions, reminders, notes/flashcards, and progress analytics into one coherent experience. The app should reduce the cognitive load of deciding what to study next and make progress visible without shaming students.

Product promise

Plan your study. Track your progress. Make your next step clear.

Core product principles

- Clarity over feature density: every screen should make the next action obvious.
- Consistency over decoration: one light theme, one spacing system, one type scale, one component library.
- Student-first and Bangladesh-aware: SSC/HSC/admission paths, local academic terms, Bangla-friendly typography, and low-bandwidth tolerance.
- Offline-tolerant: core planning and tracking should remain usable without a network connection.
- Honest progress: show what is complete, pending, overdue, or unknown; never fabricate results.
- Accessible and calm: readable contrast, large tap targets, restrained animation, and no guilt-driven streak messaging.

2. Goals, non-goals and success measures

Goals for MVP

- A student can configure their academic path and create a useful study plan in under five minutes.

- A student can add, edit, complete, reschedule and review study tasks.
- A student can track subject/chapter/topic progress and mark weak areas.
- A student can run focus sessions and see their study history.
- A student can receive local reminders for tasks and exams.
- A student can see honest weekly and subject-level progress.
- The app feels like one premium product from splash screen through settings.

Non-goals for first release

- Live classes, tutor marketplace, social feed, public profiles or chat.
- Full AI tutor, AI-generated answers, automatic question solving, or claims of guaranteed exam success.
- Payments, subscriptions, ads, parent portals or teacher dashboards.
- A comprehensive, officially verified question bank for every board and admission exam.
- Automatic timetable creation that promises perfect schedules without user confirmation.

Initial product metrics

Metric	Definition	Initial validation target (hypothesis)
Activation	New user finishes setup and creates their first task	60%+ of users who start setup
First-week usefulness	Activated users complete at least 3 tasks in first 7 days	40%+
Day-7 retention	User opens app on day 7 after first use	20%+ in private beta
Task completion	Completed tasks / due tasks, excluding deleted tasks	Observe baseline; do not game metric
Crash-free sessions	Sessions without app crash	99%+ before wider release
Qualitative satisfaction	Student says app makes planning easier	At least 8/12 interviewees in usability test

These are hypotheses for early testing, not external benchmarks or guarantees. Review them after the first 20–50 testers.

3. Target users and journeys

Primary segments

Segment	Needs	MVP behavior
SSC student	Manage school subjects, chapter coverage and exam countdown	Select SSC, board, group if applicable, exam date; track subjects and tasks
HSC student	Manage group subjects, practicals, revision and board exam prep	Select HSC, board, group, subjects; use chapter tracker and reminders
Admission candidate	Manage multiple target exams, topic revision and mock-test preparation	Select admission path and target(s); create subject/topic tasks and exam countdowns

Key user journeys

1. First launch → welcome/onboarding → academic level → academic details → exam goal → dashboard.
2. Dashboard → add task → select subject/chapter → due date/time/estimated duration → save → task appears in today's list.
3. Syllabus → open subject → open chapter → update topic status → progress recalculates immediately.
4. Dashboard or Focus tab → start timer → pause/resume/finish → save session → analytics updates.
5. Reminder notification → open the relevant task or exam screen → mark complete or reschedule.
6. Settings → change theme/notifications/account → preferences persist after restart.

4. Information architecture and navigation

Primary app navigation

Use a persistent bottom navigation bar with exactly five destinations. It should appear on the main authenticated app surfaces, not on splash, onboarding, authentication, full-screen focus mode, or modal forms.

Destination	Icon direction	Purpose
Home	House	Today's priorities, exam countdown, progress summary, quick actions
Plan	Calendar / checklist	Calendar, daily/weekly plan, task list, add/edit task
Track	Book/check	Syllabus, subjects, chapters, topic progress
Tools	Timer / grid	Focus timer, notes, flashcards, study log; tools hub
Profile	Person	Student profile, goals, preferences, settings

Header and bottom-bar placement — mandatory

- The system status bar belongs at the very top and must respect Android system insets. Never place app content underneath it.
- On standard screens, the app top app bar/header sits immediately below the status bar. It contains the page title or brand, and only context-relevant actions such as search or notifications.
- The main scrollable content begins below the app bar. It must not overlap or be hidden behind the header.
- The bottom navigation bar sits at the bottom of the app content area, above the Android gesture/navigation inset. It must not cover the last list item, primary button, or text field.
- Scrollable screens need bottom content padding at least equal to bottom navigation height plus safe-area inset. Use Scaffold or equivalent insets-aware layout rather than hard-coded screen coordinates.
- On forms, the keyboard must not cover the active field or save action; allow scrolling and resize/pan appropriately.
- On the Focus Timer screen, hide the normal bottom navigation to reduce distraction. Provide an obvious exit/back control and confirm before discarding an active session.
- Use one shared AppScaffold/AppBar/BottomNavigation implementation across all applicable screens. Do not recreate them independently per screen.
- Do not put the bottom navigation at the top. Do not duplicate the header, render two navigation bars, or place a floating bottom bar over essential content.

Navigation behavior

Use a single, predictable navigation model with back-stack preservation. Switching bottom tabs should preserve reasonable tab state (e.g., selected date or scroll position). Back should close a modal or return to the previous screen; it should not unexpectedly exit the app from nested screens. Show selected-tab state clearly.

5. Visual design system

Design direction

Premium, modern, sleek, calm, airy, polished, and purposeful. The provided reference uses a light interface with pale blue surfaces, navy text, teal actions, soft cards, thin borders, and restrained gradients. Use that as the unified direction. The collage's dark splash/end branding panels are artwork-only exceptions; all functional app screens must use the same light theme for MVP.

Theme consistency — non-negotiable

- Default theme: light. Do not alternate dark and light themes between screens.
- Dark mode may be added later only as a complete, deliberate design-system implementation with every screen, component, chart, dialog and status state covered. Do not partially introduce it in MVP.
- All screens use the same background, typography, spacing, card radius, button height, icon family and color tokens.
- Do not use random gradients, multiple competing accent colors, heavy glassmorphism, excessive shadows, or decorative elements that reduce legibility.
- The splash may use a branded illustration or gradient, but after onboarding the functional UI returns to the unified light theme.
- Use the StudyHub name exactly, with capital S and H. Do not use StudyBondhu or other placeholder names in UI, app metadata, assets, package labels or empty states.

Suggested design tokens

Token	Value / guidance
Primary brand	Deep navy #12345A
Primary action	Teal #087F8C; use a darker accessible shade if contrast requires
Accent	Soft sky blue #DCEEFF; used for selected/secondary surfaces
Background	Very light cool white #F7FAFD
Surface/card	#FFFFFF
Primary text	#172B4D
Secondary text	#64748B
Success	#168A69
Warning	#D97706
Error	#D14343
Border	#E2EAF2
Card radius	16–20 dp, consistent
Button height	48–52 dp minimum
Touch target	At least 48 × 48 dp
Spacing scale	4, 8, 12, 16, 20, 24, 32 dp
Page horizontal padding	16–20 dp on typical phones
Motion	Short, subtle transitions; typically 150–250 ms; respect reduced-motion preferences

Typography

- Use one clean sans-serif family with complete Bangla glyph coverage; verify the chosen font renders Bangla conjuncts correctly on target Android devices.
- Use a clear hierarchy: page title 22–26 sp, section title 16–18 sp, body 14–16 sp, metadata 12–13 sp. These are guidelines, not absolute rules.
- Avoid tiny text, excessive uppercase, long all-caps labels, and thin font weights on pale backgrounds.
- Support both Bangla and English strings without clipped buttons, broken line-height, or overflow. MVP may launch in English with Bangla-friendly typography; language switching should be architecture-ready.

Components to implement once and reuse

Component	Required states / variants
AppBar	Default title, back title, optional action(s), loading if needed
BottomNavigation	Five tabs; selected/unselected states; safe-area padding
PrimaryButton	Default, pressed, disabled, loading
SecondaryButton / TextButton	Default, pressed, disabled
SubjectChip	Default, selected, disabled
ProgressBar / ProgressRing	0%, partial, 100%, accessible label
TaskCard	Pending, completed, overdue, due soon
EmptyState	Illustration/icon, concise explanation, one clear CTA
FormField	Default, focused, error, disabled, helper text

ModalSheet / Dialog	Confirm, cancel, destructive confirmation
Toast / Snackbar	Success, warning, error, undo where useful
Skeleton	Loading placeholders matching final layout
DatePicker / TimePicker	Locale-aware, validation and cancel

Iconography and imagery

Use one coherent outline icon family. Keep icons the same visual weight and size. Do not mix emoji with production icons. Use illustrations sparingly, mostly for onboarding and empty states. All meaningful icons need accessible labels where the control's purpose is not obvious.

6. Screen-by-screen requirements

6.1 Splash screen

Purpose: establish brand and transition quickly to the app.

- Show StudyHub wordmark and graduation-cap/learning mark, tagline "Plan • Learn • Succeed", and a restrained branded background.
- Use the same logo asset throughout the app. Do not generate a different logo on every screen.
- Keep splash brief; no fake progress bar. Route based on persisted onboarding/authentication state.
- First-time user → onboarding. Returning signed-in user → Home. Returning local-only user → Home or last valid destination. Invalid/expired session → appropriate sign-in path without deleting local data.

6.2 Onboarding carousel

Use 2–3 concise slides, not a long marketing walkthrough. Topics: Organize your study life; Track syllabus progress; Focus and improve. Include Skip, Next and Get Started with a visible page indicator. Keep illustrations in the same style. Skip and Get Started both continue to academic setup.

6.3 Academic level selection

Selectable cards: SSC, HSC, Admission. Each card has one icon, a clear label and one-line description. Selected state uses teal border/fill accent and a check indicator; selection is not indicated by color alone. Continue stays disabled until a valid choice is made.

6.4 Academic profile setup

Collect only what is needed to tailor the app. Fields vary by path: name/nickname (optional), level, board (where applicable), group/stream, subject selection, target exam(s), exam date if known, daily available study time, preferred reminder time. Allow 'I'm not sure' or skip for nonessential fields. Do not force a phone number or date of birth for a local-only MVP. Validate date and required choices.

6.5 Account / local mode

Prefer a friction-light entry. The user should be able to explore core planning locally without being forced to register immediately. If cloud backup is enabled, provide sign-in/sign-up and explain why it is useful. Account creation, password reset and sign-out must be functional if shown. Never show decorative Google sign-in unless it is wired to a correctly configured provider.

6.6 Home dashboard

Header: greeting and student display name; right-side notification action only if notification screen exists. The top app bar must align consistently with all other standard screens.

- Exam countdown card: target exam name, days left, date. If no date is set, show a setup CTA rather than a fabricated countdown.
- Syllabus progress summary: percent based on completed tracked items divided by total tracked items. Empty syllabus = “Set up your syllabus”, not 0% framed as failure.
- Today’s plan: tasks sorted by due time/priority; show subject, short task title, estimated duration and status.
- Quick actions: Add task, Start focus, Track syllabus. Limit to 2–3 prominent actions.
- Daily study summary: time studied today compared with optional personal target.
- A supportive empty state for new users: create first task or choose subjects.
- No fake student name, exam date, streak, marks or progress in production. Sample/demo mode must be explicitly labelled.

6.7 Study Plan / Calendar

- Offer Day and Week views for MVP; Month view can be a compact calendar if straightforward.
- Date selector has clear selected/current-day states and supports previous/next navigation.
- Task list shows title, subject, estimated time, due time, completion state and overdue marker.
- Add Task button remains easy to find but does not obscure list content.
- Filters: All, Pending, Completed; avoid excessive filters in MVP.
- Empty day shows CTA to add a task or move to another date.
- Changing dates must refresh the visible list; timezone/date calculations use device local time.

6.8 Add / Edit Task

Fields: task title (required), subject (optional only if user has no subjects configured), chapter/topic (optional), date (required with sensible default), start time (optional), estimated duration, priority (Normal/High/Low), notes (optional), reminder toggle (optional).

- Primary CTA: Save Task. Secondary: Cancel/back.
- Validate title and date; prevent accidental duplicate taps while saving.
- Edit mode prepopulates every saved field.
- Completion is separate from editing. Completed tasks remain in history and can be reopened if needed.
- Deleting requires confirmation; offer undo when possible.
- On save, return to the prior context and show the task immediately. Do not silently lose data on validation errors.

6.9 Syllabus Tracker

- Tabs or filters for the user’s configured academic path; never display all levels as though they all apply to one student.
- Subject list displays subject name, completed/total chapters, progress bar and optional weak-area marker.
- Chapter states: Not started, In progress, Completed. Use text/icon as well as color.
- Progress calculation: completed items / total items × 100, with a documented unit. Prefer topic-level progress if all topics exist; otherwise use chapter-level progress. Do not mix denominators silently.
- Allow custom subjects/chapters if official curriculum data is missing or the student’s curriculum differs.
- Keep syllabus content data-driven and versioned by level, board, group, curriculum year and source/verification status.
- Clearly mark built-in syllabus as verified only after review against official source material. Otherwise label it editable/sample.

6.10 Chapter details

Show chapter name, subject, progress, topic checklist, status actions and optional notes/resource links. Users can mark individual topics complete and set a topic as weak/revisit. Provide a simple “Mark chapter complete” action that updates topic states only after confirmation if it would overwrite incomplete topic detail. Do not show unverified textbook pages or resource links as official.

6.11 Focus Timer

- Default 25-minute focus / 5-minute break, plus Custom mode. Allow durations to be configured within sensible bounds.
- Large, legible timer ring with remaining time, Start/Pause/Resume and End controls.
- Optional subject/task association before starting.
- When app goes to background, timer state must remain accurate using timestamps; do not rely only on a decrementing UI counter.
- On finish, show a session summary and ask to save/discard if required. Avoid duplicate session records.
- If a session is interrupted by process death, recover gracefully and let the user resume or discard.
- Hide bottom navigation in focus mode; respect system status/navigation bars and accessibility font sizes.

6.12 Study session log

Show daily/weekly/monthly summaries, total focused time, session count and subject distribution. Sessions must be derived from saved session records, not hard-coded sample values. Explain whether paused time is excluded. Allow a user to remove an erroneous session with confirmation.

6.13 Performance analytics

MVP analytics should focus on effort and planning reliability, not pretend to predict exam marks. Show weekly study time, task completion, syllabus progress and subject-wise study time. If test results are supported, make them manually entered and clearly labelled. Every chart needs a title, units, date range, accessible text summary and empty state. Use consistent chart colors across the app.

6.14 Notes and flashcards

Notes: create, edit, delete, search by title/content, associate with subject/chapter, save timestamp. Flashcards: create front/back, flip/reveal, navigate, optionally mark “Know it” / “Review again”. MVP stores content locally and syncs if cloud mode is enabled. Rich text formatting is not required; plain text is acceptable. Never imply that notes are backed up unless sync is working.

6.15 Reminders and notifications

- Categories: study task, revision, exam countdown/event.
- Ask for notification permission contextually and explain the benefit before the system prompt.
- Allow per-reminder enable/disable, time edit and deletion.
- Notification tap opens the relevant task or exam detail, not always Home.
- If exact alarms are restricted by Android policy/device settings, use an appropriate supported scheduling mechanism and communicate limitations.
- Handle permission denied without blocking core app usage.

6.16 Profile

Show display name, academic path, group/board where applicable, exam goal/date, study summary and edit profile action. Do not expose private email/phone publicly. Allow user to update preferences without resetting tasks or progress.

6.17 Settings

MVP sections: appearance (light only; theme setting may be omitted until dark mode is complete), notification preferences, study preferences, data backup/sync status, privacy, help/about, sign out, and account deletion if accounts exist. Do not show nonfunctional switches. Provide app version and contact/support route when available.

6.18 Search, loading, empty and error states

If global search is not in MVP, do not show a search icon that does nothing. All async operations need loading, success, empty and error states. Use concise, actionable errors and preserve user-entered form data after failure. Empty states should explain what the user can do next.

7. Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria
FR-01	First launch and onboarding	First-time state shows onboarding; Skip/Get Started reaches level setup; returning user is not forced through onboarding again.
FR-02	Academic setup	Level and required profile selections persist after app restart; conditional fields match selected path.
FR-03	Task CRUD	User can create, view, edit, complete, reopen and delete tasks; all changes persist.
FR-04	Date filtering	Tasks shown correspond to selected date and filters; timezone uses device local time.
FR-05	Syllabus progress	Changing chapter/topic state recalculates progress correctly and updates subject/dashboard summaries.
FR-06	Timer	Start/pause/resume/end work; elapsed time remains accurate across backgrounding; a finished session is not duplicated.
FR-07	Reminders	Permission granted → scheduled reminder fires approximately as configured; tap routes to related item; denied permission is handled gracefully.
FR-08	Analytics	All values derive from saved user records; no fabricated demo values in production.
FR-09	Notes/flashcards	Create/edit/delete/search work and data persists.
FR-10	Navigation	Bottom bar appears only on main app surfaces, with correct safe areas; no content is obscured.
FR-11	Offline	Core task/syllabus/note operations remain usable offline for local records; network failure does not erase work.
FR-12	Privacy	A user cannot read or modify another user's cloud records; deletion and sign-out behavior are documented.
FR-13	Visual consistency	All functional screens use the same light design tokens and shared components; no accidental dark screens.
FR-14	Accessibility	Touch targets are generally ≥48dp; content supports text scaling; icons/controls have meaningful labels.

8. Data model

Use a relational model with stable IDs, timestamps, ownership and clear links. Exact table names can be adjusted by the implementation, but relationships and privacy boundaries must remain.

Entity	Suggested fields	Notes
profiles	id, display_name, level, board, group_name, daily_goal_minutes, preferred_reminder_time, created_at, updated_at	id links to authenticated user when cloud account exists; avoid unnecessary personal data
academic_goals	id, user_id, goal_type, target_name, exam_date, status	Allows multiple admission targets later; exam_date nullable
subjects	id, user_id, curriculum_subject_id nullable, name, color_token, sort_order, is_active	User-defined and built-in subjects both supported
chapters	id, subject_id, curriculum_chapter_id nullable, title, status, sort_order, updated_at	Status enum; curriculum link optional
topics	id, chapter_id, title, status, is_weak, sort_order, updated_at	Optional; progress denominator must be consistent
study_tasks	id, user_id, subject_id nullable, chapter_id nullable, title, notes, due_at, estimated_minutes, priority, status, created_at, updated_at, completed_at	Use explicit timezone strategy; local display in device timezone
study_sessions	id, user_id, task_id nullable, subject_id nullable, started_at, ended_at, focus_seconds, status, created_at	Store timestamps; focus_seconds excludes paused periods
notes	id, user_id, subject_id nullable, chapter_id nullable, title, body, created_at, updated_at	Searchable text; define sync behavior
flashcards	id, user_id, subject_id nullable, chapter_id nullable, front_text, back_text, review_state, created_at, updated_at	Keep review state simple for MVP
reminders	id, user_id, task_id nullable, goal_id nullable, scheduled_at, enabled, reminder_type, created_at, updated_at	Local notification IDs must be mapped to records
curriculum_versions	id, level, board, group_name, curriculum_year, source_url, verification_status, updated_at	Admin/content workflow can be manual for MVP

Data and sync rules

- Generate stable unique IDs; never use visible list index as a database identifier.
- Use created_at/updated_at consistently. Soft-delete only where needed; otherwise delete intentionally with confirmation.
- If offline-first local storage is implemented, define a sync queue and conflict policy before enabling multi-device sync.
- For conflicting edits, prefer updated timestamps only as a basic strategy; do not silently overwrite user data. A simple “last write wins” policy must be documented and tested if used.
- Use database migrations for schema changes. Do not create or modify production schema from ad-hoc client-side logic.
- Never store secrets, service-role keys or privileged backend credentials in the Android app.

9. Technical architecture for Google AI Studio

Recommended baseline

Layer	Recommendation	Implementation notes
Android UI	Flutter + Dart, if building native-style Android-first app	Use shared widgets, responsive layouts, app navigation and a clear feature-based folder structure.
Local persistence	SQLite-backed local database or an established Flutter local persistence	Core tasks, syllabus state, notes and sessions should survive app restart and

	package	temporary offline use.
Backend	Supabase Auth + PostgreSQL + Row Level Security, if cloud sync is included	Use migrations, per-user ownership policies and least-privilege access.
Notifications	Flutter local notifications package or equivalent maintained solution	Test Android permission behavior, exact-alarm limitations and deep links.
Monitoring	Crash reporting and basic analytics with privacy-conscious configuration	Do not collect note contents or unnecessary personal data.
Design	Figma or equivalent source of truth + reusable design tokens	Implement shared components, not per-screen styling drift.

Important tool constraint: Google AI Studio can assist with planning and generating code, but verify its current project/export capabilities and the selected runtime before committing to a workflow. If using AI Studio's web-app generation path, do not assume it automatically produces a production-grade native Android app. Confirm whether the generated output is Flutter/native Android or a web app packaged for Android, and test platform capabilities accordingly.

Suggested project organization

- lib/app/ — app bootstrap, router, theme, design tokens
- lib/core/ — date/time utilities, validation, error handling, shared types
- lib/shared/widgets/ — buttons, cards, fields, progress indicators, app bar, bottom navigation
- lib/features/onboarding/ — onboarding and academic setup
- lib/features/home/ — dashboard and summary cards
- lib/features/planner/ — calendar, tasks, task editor
- lib/features/syllabus/ — subjects, chapters, topics, progress
- lib/features/focus/ — timer, session summary and log
- lib/features/analytics/ — weekly and subject-level summaries
- lib/features/notes/ — notes and flashcards
- lib/features/reminders/ — scheduling and notification routing
- lib/features/profile/ — profile and settings
- lib/data/local/ — local database, DAOs/repositories
- lib/data/remote/ — Supabase client, remote repositories, sync
- test/ — unit, widget and integration tests

Engineering rules

- Separate UI, business logic and data access. Widgets should not directly embed complex database logic.
- Use a repository/service boundary so local storage and remote sync can evolve independently.
- Use a single source of truth for date/time handling, progress calculation and design tokens.
- Add unit tests for progress calculations, due-date filtering, timer duration and task completion.
- Use typed models and explicit status enums rather than arbitrary strings scattered through the UI.
- Do not put secrets or privileged API keys in source control or generated client code.
- Pin and review package versions; use maintained packages and inspect permissions they request.

10. Google AI Studio implementation instructions

Paste this PRD into the project context and upload the StudyHub reference montage. Work in small milestones. Do not ask the coding assistant to build every screen and backend in one giant generation, because that often creates inconsistent navigation, duplicate components and fake interactions.

Master instruction for the coding assistant

You are the senior product designer and senior engineer implementing StudyHub, an Android-first study management app for Bangladesh's SSC, HSC and university admission students. Treat the attached StudyHub master UI montage as the visual direction and screen inventory, and treat this PRD as the authoritative behavior specification.

NON-NEGOTIABLE UI RULES:
1. Use the StudyHub name exactly.
2. Use one consistent light theme on all functional screens. Do not alternate light and dark screens. Dark branding artwork in the reference is not a dark-mode requirement.
3. Respect Android status-bar, gesture-bar, keyboard and safe-area insets.
4. On standard screens, place the app header immediately below the system status bar and the persistent five-item bottom navigation at the bottom, above the system navigation inset. Main content belongs between them. Add enough bottom padding so content is never covered.
5. Hide bottom navigation only for onboarding/authentication, modal flows where appropriate, and the immersive Focus Timer. Never place the bottom bar at the top.
6. Implement the header and bottom navigation once as reusable shared components. Use one spacing scale, type scale, color palette, card radius, button height and icon family.
7. Premium means clean hierarchy, generous whitespace, readable typography, subtle borders/shadows, restrained teal/blue accents, smooth but purposeful transitions, and accessible contrast—not lots of gradients or decoration.
8. Do not invent data. Use clearly labelled demo seed data only in demo/development mode. Production screens must show actual persisted data or helpful empty states.
9. Every visible primary control must work. No fake buttons, dead search icons, fake Google login, placeholder notification settings or static progress values presented as real.
10. Build incrementally, run the app/tests after each milestone, fix compilation/runtime errors, and preserve working features.

IMPLEMENTATION APPROACH:
First inspect the current project and report its actual framework/runtime. Do not assume a web app is a native Android app. Then propose a brief architecture and implement Milestone 1 only: app shell, design tokens, navigation, splash/onboarding and academic setup. Wait for validation before moving to the next milestone. Keep UI, business logic and data repositories separate. Follow all functional requirements, data rules, accessibility requirements and acceptance criteria in the PRD. If any requirement is ambiguous or unsupported by the chosen runtime, state the limitation and propose the closest reliable implementation rather than silently omitting it.

Recommended build milestones / prompts

Milestone	Implement	Done when
M1 — Design foundation	Theme tokens, fonts, shared components, splash, onboarding, academic setup, app shell	All screens align with reference; one light theme; navigation/insets correct
M2 — Local data	Models, local database, repositories, sample data isolated to debug mode	Data survives restart; no production fake values
M3 — Planner	Home, calendar/day/week views, task create/edit/complete/delete	Task lifecycle and date filters pass tests
M4 — Syllabus	Subject/chapter/topic views, progress math, weak-area marker	Progress is consistent and updates across screens
M5 — Focus and logs	Timer state machine, session save, history	Background timer accurate; no duplicate sessions
M6 — Notes and reminders	CRUD/search notes, flashcards, local notifications	Permission and error cases handled
M7 — Analytics/profile	Derived metrics, profile/settings, backup/sync status	All metrics use actual data
M8 — QA/release	Accessibility, low-end device, offline, privacy, store assets	Release checklist passes

Prompt template for each milestone

“Implement only Milestone [X] from the StudyHub PRD. Before coding, list files to create/change and explain any assumptions. Reuse existing design tokens and shared components. Do not redesign other screens or break existing features. Implement loading, empty, success and error states. Add tests for relevant business logic. Run/build/test if supported and report exact results, remaining limitations, and manual verification steps. Do not claim a test passed unless it actually ran.”

11. Privacy, security and student safety

- Collect the minimum data needed. Avoid collecting date of birth, school address, precise location or contacts for the MVP.
- If users may be minors, provide age-appropriate privacy language and review applicable Bangladesh laws and Google Play policies before release.
- Keep user records private using backend access policies. Test with two separate accounts to ensure isolation.
- Do not log task notes, note contents, passwords, access tokens or other sensitive fields in analytics/crash logs.
- Use secure authentication flows and store session tokens using platform-appropriate secure storage.
- Provide a clear way to export or delete user data where required. Account deletion must also address associated cloud data and explain what happens to local data.
- Use official curriculum sources only with appropriate rights and attribution. Do not scrape or republish copyrighted textbooks, coaching materials or question banks without permission.
- Avoid manipulative streaks, shame-based copy, guaranteed-result claims or messages that imply the student has failed as a person.

12. Performance, reliability and accessibility

Area	Requirement
Launch	Show useful UI quickly; avoid unnecessary splash delay and blocking network calls.
Scrolling	Lists remain smooth on entry-level Android devices; paginate or virtualize large lists.
Network	Show offline/sync state when relevant; retry safely without duplicating writes.
Data safety	No lost task/note when an operation fails; preserve draft input.
Text scaling	Support Android font scaling without clipped text or inaccessible controls.
Contrast	Meet WCAG-inspired contrast targets where practical; verify actual text/background pairs.
Touch	Controls generally at least 48×48 dp; adequate spacing for one-handed use.
Motion	Avoid excessive motion; respect system reduced-animation settings where available.
Localization	Use resource-based strings; support English now and Bangla localization without hard-coded UI strings.
Battery	Timer and reminders must not use continuous high-frequency background work unnecessarily.

13. Testing strategy

Unit tests

- Syllabus percentage with zero, partial and complete items.
- Task filtering by date, status and overdue state.
- Timezone/day-boundary behavior.
- Focus timer start, pause, resume, finish, interruption and duplicate prevention.
- Weekly study totals and subject aggregation.
- Validation for task title, dates and duration limits.

Widget / UI tests

- Onboarding Skip and Get Started paths.
- Academic level selection and conditional profile fields.
- Bottom navigation selected state and safe-area placement.
- Task creation/editing validation and success state.
- Empty, loading and error states.
- Large font size, long Bangla strings and narrow-screen overflow.

Manual device tests

- At least one low-end Android device and one newer device.
- Offline mode, slow network and interrupted requests.
- App backgrounded/force-stopped during a focus session.
- Notification permission allowed and denied.
- Keyboard visible while editing a task.
- Portrait orientation and Android system font scaling.
- Fresh install, app update, sign-out/sign-in and data deletion.

14. Release checklist

- ☐ All core MVP flows pass acceptance criteria.
- ☐ No screen uses a different theme accidentally; shared header and bottom navigation are consistent.
- ☐ No dead buttons, misleading progress, fake notifications or fake sign-in.
- ☐ Correct app name StudyHub across UI, app metadata and assets.
- ☐ No secret keys in app bundle or repository.
- ☐ Privacy policy, data safety declarations, support contact and account deletion flow are ready where required.
- ☐ Curriculum source/version and verification status are documented.
- ☐ Store listing, icon, screenshots and release signing are ready.
- ☐ Crash monitoring and basic privacy-conscious analytics are configured.
- ☐ Beta feedback reviewed and high-severity bugs resolved.

15. Risks and mitigations

Risk	Mitigation
Scope creep	Freeze MVP list; defer AI tutor, social, payments and large content libraries.
AI-generated inconsistent UI	Enforce shared components/tokens; implement one milestone at

	a time; compare every screen against reference.
Wrong curriculum content	Use versioned, sourced data; allow custom topics; never label unverified content official.
Offline/cloud conflicts	Start with reliable local persistence; add explicit sync and conflict rules before multi-device claims.
Notification unreliability	Test Android permission and scheduling restrictions on real devices; provide in-app task list as fallback.
Student privacy/minors	Minimize data, secure records, review laws and store policies before launch.
AI Studio runtime mismatch	Confirm generated project framework and Android deployment path at the beginning.

16. Decisions to finalize before implementation

Decision	Recommended MVP default
Brand	StudyHub
Theme	Light only
Navigation	Five-item bottom bar: Home, Plan, Track, Tools, Profile
Languages	English-first UI with Bangla-ready typography/localization architecture
Account requirement	Allow local use; offer account for backup/sync if supported
Syllabus data	Start with a limited, reviewed dataset for the pilot level(s); permit custom subjects/topics
Cloud sync	Optional for first local MVP; add only with tested auth, RLS and conflict handling
AI features	Out of MVP; do not block launch on them
Release strategy	Private beta with 10–20 students, iterate, then wider release

17. Final product definition

StudyHub MVP is complete when a student can install the Android app, set up an academic path, create and manage daily study tasks, track syllabus progress, run focus sessions, save notes, receive supported reminders, and view accurate progress—all through a cohesive, premium light-theme UI. Core data must persist, key flows must work offline where specified, and no screen may be merely decorative while implying a feature is functional.

End of PRD